

HOSSAM EMAD MOHAMED OSMAN

Embedded Systems / Hardware Engineer

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PROFESSIONAL SUMMARY

Embedded Systems Engineer with strong hands-on experience in **hardware design, PCB layout, firmware development, and production support**. Proven ability to analyze field issues, support deployed products, and deliver **improved hardware and firmware revisions**. Experienced in **automotive and industrial IoT systems, 4-layer PCB design, controlled impedance routing, BMS, and RTOS-based firmware**, with full ownership from design to manufacturing and deployment.

TECHNICAL SKILLS

Firmware & Software: C, Python, RTOS, SMS, MQTT, HTTP, GPS, FOTA, Geofencing, Embedded Debugging

Hardware & PCB: 4-layer PCB design, stackup planning, controlled impedance routing (USB, CAN, RF), RF basics & antenna matching, power electronics, BMS, board bring-up

Tools: Altium Designer, EAGLE, EasyEDA Pro, KiCad, Fusion 360, JLCPCB workflow

PROFESSIONAL EXPERIENCE

Embedded Systems Engineer Jul 2024 – Present

SEITech Solutions (Acquired by T&S Group) — Smart Village

SEIFence — IoT Tracking & Control Device

- Joined during fabrication and early deployment of **Hardware Version 1**, identifying hardware, mechanical, and firmware issues.
- Provided **aftersales hardware maintenance and debugging** for approximately **200 deployed units**.
- Independently proposed and developed an improved **Hardware & Software Version 2** while supporting fielded products.
- Implemented system-level enhancements resulting in **cost reduction and performance improvement**.
- Resolved **GPS antenna mechanical stability issues** by applying post-fabrication PCB fixes (drilled mounting holes and added through-hole fixation pads).
- Successfully transitioned production to Version 2 and deployed it across **120 manufactured units**.
- Firmware (sole responsibility):** designed and implemented the complete product firmware stack, including SMS, MQTT, HTTP communication, GPS data processing, FOTA, and a geofencing algorithm.
- Built firmware using an **RTOS-based architecture**; developed the full system twice — an initial **Python-based** version for validation, followed by the final **optimized production firmware in C**.

Tools: EAGLE PCB, Fusion 360

ForWheelz — Automotive Telematics Device

- Owned the complete hardware development lifecycle of an automotive telematics product.
- Selected mechanical enclosure and coordinated with suppliers for specifications and quotations.
- Architected a **two-board stacked design using two 4-layer PCBs** due to strict mechanical constraints.
- Defined PCB stackup with **outer signal/power layers and internal solid GND planes**.
- Designed STM32-based system with **three CAN interfaces (HS CAN, CAN scanning, SWCAN)**, USB, RF, and power management.
- Applied **controlled-impedance routing** for USB, CAN, and RF signals.
- Designed RF subsystem including **GSM antenna placement and impedance matching network**.
- Prepared full PCB and PCBA manufacturing files and coordinated fabrication with **JLCPCB**.

Tools: EasyEDA Pro

SEITech Box — Industrial IoT Gateway & Power Backup System

3-Cell BMS & Backup Power Module

- Designed a **3-cell BMS** to provide backup power for the Edge Box ESP100 during inverter power loss.
- Implemented controlled DC-DC power path, charging/discharging monitoring, and system validation.
- Selected enclosure, designed schematic and PCB, prepared fabrication files, and ordered PCB.
- Performed **manual assembly, soldering, board bring-up, debugging, and final validation**.
- Delivered validated product to the customer.

Tools: KiCad

Edge Box ESP100 — Reverse Engineering & Redesign

- Reverse engineered existing hardware and redesigned an **ESP32-S3–based industrial IoT gateway**.
- Integrated Wi-Fi, BLE, **SIMCom 4G**, GPS, RS232, RS485, BMS, ADC, GPIO, and power management circuits.
- Designed and validated PCB for reliable industrial deployment.

Access Control Board — Industrial Ethernet & RS485 Remote I/O Controller

Role: Hardware Design Engineer. Designed and developed a production-ready embedded controller module for remote monitoring, industrial automation, and networked control applications, built around an STM32F401 Cortex-M4 microcontroller with Ethernet, RS485, USB, EEPROM storage, digital I/O expansion, relay control, and comprehensive power management.

- Designed the complete hardware architecture based on the **STM32F401RCT6** (84 MHz ARM Cortex-M4) microcontroller.
- Implemented Ethernet connectivity using the **ENC28J60** SPI Ethernet controller with RJ45 interface for remote monitoring and control.
- Developed an isolated RS485 communication interface using **MAX3485** transceivers and optocoupler isolation for industrial environments.
- Integrated dual **24LC512** EEPROM devices over I²C for non-volatile data logging and configuration storage.
- Designed USB Type-C programming and debugging interface with independent power support for firmware updates without external system power.
- Developed a multi-rail power architecture: 12V input supply, 5V rail (LM2596S switching regulator), 3.3V rail (AMS1117 LDO).
- Implemented extensive protection circuitry: TVS transient suppression, ESD protection, over-voltage and surge protection, bulk and local decoupling networks.
- Designed digital input monitoring and solid-state output control interfaces; implemented relay-driving circuitry with electrical isolation.
- Added comprehensive system diagnostics through multiple status LEDs and hardware test points.
- Performed schematic capture, component selection, PCB layout review, BOM preparation, manufacturing documentation generation, and production support.
- Generated complete production package: Gerber files, pick-and-place files, assembly drawings, BOM with LCSC sourcing information, and revision-controlled design archives.

Tools: STM32F401RCT6, ENC28J60, MAX3485, 24LC512, USB Type-C, LM2596S, AMS1117, I²C/SPI/UART/RS485/Ethernet, KiCad PCB Design Suite

Smart Wi-Fi Switch — ESP8266-Based IoT Smart Home Device

Designed and developed a **3-channel Smart Wi-Fi Switch** that enables remote control of AC loads using an **ESP8266 (ESP-12E)**. The system integrates an AC-to-DC power supply, a 3.3V regulation stage, and relay outputs to safely switch mains-powered devices, with the PCB designed for proper isolation between the high-voltage AC section and low-voltage control circuitry.

Key Features:

- 3-channel AC load control using electromechanical relays.
- ESP8266-based Wi-Fi controller with UART programming interface.
- Integrated AC-to-5V power supply with 3.3V regulation for logic circuitry.
- Safe PCB routing for both AC mains and low-voltage signals.
- Compact hardware designed for embedded IoT applications.

My Contributions:

- Defined the system architecture and selected hardware components.
- Built and validated the initial prototype on an FR4 prototype board.
- Designed the complete schematic and PCB layout using **KiCad**.

- Generated manufacturing files (Gerber/BOM) for PCB fabrication.
- Assembled the PCB and programmed the ESP8266 firmware via UART.
- Performed hardware debugging, functional validation, and end-to-end system testing.

Tools: ESP8266 (ESP-12E), KiCad, Embedded C, UART, PCB Design, AC-DC Power Supply, Relay Control, IoT Hardware Development

Industrial IoT Controller & Hardware Development Platform

Designed and developed a custom Industrial IoT Controller based on the STM32F407 and ESP32-S3, integrating industrial communication interfaces, wireless connectivity, external memory, environmental sensing, and power management into a single hardware platform. The project was developed from concept through PCB fabrication, assembly, firmware flashing, and hardware validation.

Key Features:

- STM32F407 main controller with ESP32-S3 wireless coprocessor.
- Ethernet connectivity using ENC28J60.
- CAN Bus interface using TJA1044 transceiver.
- RS-485 communication with logic-level translation and surge protection.
- 512 Mbit Quad SPI Flash (W25Q512) for external storage.
- Environmental monitoring using BME280 temperature, humidity, and pressure sensor.
- Motion detection using LIS3DH accelerometer.
- Ambient light sensing (LDR) and analog voltage monitoring.
- USB Type-C interface for programming and debugging.
- Dedicated SWD/JTAG and UART programming interfaces.
- Multi-stage power supply generating regulated 5 V and 3.3 V rails from a 12 V input.
- Multiple expansion headers for GPIO and peripheral interfaces.

My Contributions:

- Defined the complete hardware architecture and component selection.
- Designed the multi-sheet schematic and PCB layout using KiCad.
- Implemented high-speed interfaces (QSPI, SPI, Ethernet, USB, CAN, and RS-485).
- Designed the power distribution and protection circuitry.
- Performed PCB fabrication review, assembly, and hardware bring-up.
- Programmed and validated both STM32F407 and ESP32-S3.
- Conducted functional testing, debugging, and hardware verification.

Tools & Technologies

STM32F407 • ESP32-S3 • KiCad • Embedded C • Ethernet (ENC28J60) • CAN • RS-485 • QSPI Flash • USB Type-C • SWD/JTAG • PCB Design • Hardware Bring-up • Industrial IoT

Embedded Systems Engineer Jan 2024 – Feb 2024

Noziqs Medical — Nasr City

- Developed and deployed a generic control board for operating tables, controlling the solenoid valves for table motion, the hydraulic pump on/off, and battery charging management.
- Deployed the solution across **12 operating tables**.
- Developed embedded firmware for operating tables, improving system functionality by **30%**.
- Reduced device downtime by **20%** through board-level troubleshooting and maintenance.

EDUCATION

Bachelor of Electronics and Control Engineering *Sep 2017 – Jul 2022*

Menoufia University — Grade: Very Good

DIPLOMAS & PROFESSIONAL TRAINING

Automotive Embedded Systems Diploma

National Telecommunication Institute (NTI) — New Capital | Feb 2024 – Jul 2024

- **Applied Technical Program (288 hours):** Embedded C, RTOS, automotive software architecture, STM32 & AVR microcontrollers, CAN, LIN, UART, SPI, I2C, bootloader and **FOTA**, automotive testing (unit, integration, validation).
- **Soft Skills & Project Management (132 hours).**
- Completed multiple **ADAS projects**.

Crash Course Electronics and PCB Design — Udemy

Instructor: Andre LaMothe | Completed: Jan 2026 | 112 hours

- Electronics fundamentals, AC/DC analysis, analog & digital electronics.
- Full PCB design flow: schematic, layout, fabrication, assembly, bring-up, and debugging.

Hardware Design Specialization Mentorship (Ongoing)

- Professional PCB design using **Altium Designer**.
- Designed **ESP32-S3** and **RP2040 development boards**.
- Focus on professional schematics, **proper grounding**, and advanced PCB layout practices.

SELECTED ACADEMIC & TRAINING PROJECTS

NTI Project — Advanced Driver-Assistance System (ADAS)

Apr 2024 – May 2024

- Developed ADAS solution with **ACC**, **LKA** using **FreeRTOS**.
- Implemented BLE-based vehicle control, GPS tracking, emergency mode, and **FOTA**.

Advanced Driver Assistance System (AVR-Based)

Apr 2024

- Implemented ACC, BAS, SPLS, DM, HUD with LCD interface and peripheral drivers.

Portable Hemodialysis Machine

Apr 2019

- Designed hardware and software for a portable hemodialysis machine, increasing treatment accessibility by **50%**.

Graduation Project — Application of Virtual Reality in Healthcare

Team Leader

- Led a team of 6 engineers; achieved **2nd place** in a scientific exhibition.

LANGUAGE

English